

Name: _____

Date: _____

PROPORTION, (INVERSE PROPORTION)

LO: I can solve problems involving inverse proportion.

Inverse proportion

Two quantities are inversely proportional if when one doubles, the other halves. Their product stays constant.

Formula: $y = \frac{k}{x}$ where k is constant. As x increases, y decreases.

Quick examples

●	$y = \frac{k}{x}$ means y is inversely proportional to x	formula
●	If 4 workers take 6 hours, 2 workers take 12 hours	inverse
●	$x \times y = k$ (constant product)	defining property
●	Inverse proportion means $y = \frac{k}{x}$	that is direct

Key idea

In inverse proportion, the product xy is always constant. $y = \frac{k}{x}$.

$$y = \frac{k}{x}$$

$k = x \times y$ (constant product), x means y

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - workers and time

4 builders take 15 days to build a wall. How long for 6 builders?

$$k = 4 \times 15 = 60 \text{ (worker-days)}$$

$$\begin{aligned} 6 \text{ builders: time} &= 60 \div 6 \\ &= 10 \text{ days} \end{aligned}$$

$$\boxed{= 10 \text{ days}}$$

More workers means less time: inverse proportion.

Example 2 - finding k

y is inversely proportional to x . When $x = 3$, $y = 8$. Find y when $x = 6$.

$$k = x \times y = 3 \times 8 = 24$$

$$y = 24 \div x$$

$$\text{When } x = 6: y = 24 \div 6 = 4$$

$$\boxed{y = 4}$$

Find the constant product, then divide.

Example 3 - speed and time

At 60 mph, a journey takes 2 hours. How long at 40 mph?

$$\begin{aligned} k &= \text{speed} \times \text{time} = 60 \times 2 = 120 \\ &\text{(distance)} \end{aligned}$$

$$\begin{aligned} \text{At 40 mph: time} &= 120 \div 40 \\ &= 3 \text{ hours} \end{aligned}$$

$$\boxed{= 3 \text{ hours}}$$

Speed and time are inversely proportional for a fixed distance.

Example 4 - inverse square

y is inversely proportional to x^2 . When $x = 2$, $y = 5$. Find y when $x = 4$.

$$\begin{aligned} y &= \frac{k}{x^2}. \quad k = y \times x^2 = 5 \times 4 = 20 \\ y &= 20/x^2 \end{aligned}$$

$$\text{When } x = 4: y = \frac{20}{4^2} = 1.25$$

$$\boxed{y = 1.25}$$

Inversely proportional to x^2 means $y = \frac{k}{x^2}$.

YOUR TURN – PRACTICE (deliberate practice)

1. Workers and time

Find the constant, then solve.

- a) 3 workers take 12 hours. Time for 4 workers?
- b) 5 machines take 8 hours. Time for 10 machines?
- c) 6 people take 5 days. Time for 2 people?
- d) 8 taps fill a pool in 3 hours. Time for 6 taps?

2. Finding k and using the formula

y is inversely proportional to x. Find the missing value.

- a) $x = 4$, $y = 10$. Find y when $x = 8$
- b) $x = 5$, $y = 12$. Find y when $x = 3$
- c) $x = 2$, $y = 18$. Find x when $y = 6$
- d) $x = 6$, $y = 4$. Find y when $x = 12$

3. Speed and time

Distance is fixed. Find the missing value.

- a) At 30 mph takes 4 hours. Time at 60 mph?
- b) At 50 km/h takes 3 hours. Speed to take 2 hours?
- c) At 80 km/h takes 1.5 hours. Time at 40 km/h?
- d) At 45 mph takes 2 hours. Time at 90 mph?

4. Word problems

Identify inverse proportion and solve.

- a) 12 sweets shared among 4 children gives 3 each. Shared among 6?
- b) A food supply lasts 10 days for 6 people. How long for 4 people?
- c) Pressure doubles when volume halves (gas law). Vol = 8 at P = 3. Vol at P = 6?
- d) A pipe fills a tank in 12 min. 3 identical pipes fill it in?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) In inverse proportion, $xy = \text{constant}$ ☐

Correct answer: _____

Reason: _____

b) Inverse proportion means $y = kx$ ☐

Correct answer: _____

Reason: _____

c) More workers means less time (inverse proportion) ☐

Correct answer: _____

Reason: _____

d) The graph of $y = k/x$ is a straight line ☐

Correct answer: _____

Reason: _____

e) If x doubles, y halves in inverse proportion ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) 8 workers can paint a school in 6 days. The head teacher wants it done in 4 days. How many workers are needed? If each worker costs £120 per day, what is the total cost for 6 days vs 4 days?

Expression: _____

Simplified: _____

b) y is inversely proportional to x. When $x = 4$, $y = 9$. Find: (i) the formula, (ii) y when $x = 16$, (iii) x when $y = 6$.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

